Chapter 5

Document Design and Graphics

Designing the Document: Document design is the knots and bolts of technical writing. No matter how brilliant or important the content, if it is not formatted in the way that enhances reliability, it will likely not receive the attention it deserves. So, you need to understand the importance of reliability to your technical audience, and what that looks like in technical document. You need to know how to use heading to organize information logically to enhance readers attention. Understanding the rules for embedding various kinds of list in your document to emphasize key points and simplify text. Understand how to integrate various kinds of figures and tables into documents to effectively present visual data and images. Revision strategies to enhance clarity and reliability are the factors needed to be considered while designing the document.

So, designing the document should be done perfectly. A cluttered room with poorly designed lighting can make it difficult for people to find what they are looking for. Similarly, a clutter document with poorly designed elements can put readers in the same predicament. Readers may be discouraged if they cannot find the information they need quickly and easily. When you write, you make many decisions- what to write, how to organize which words to use, and so on. Indeed, words are important to any writer, but in technical writing, how the words look on the page is just as important as what the word say. If they want readers to stay focused, good technical writers most learn to design pages that are visually friendly. Without proper documentation, customers are in the dark about the purpose of your writing. It becomes hard to point out the real issues.

Document design helps audience understand or document through visual elements. Too small prints and small font size refers to the poorly designed document. Larger prints and larger font sizes should be designed with the readers in mind.

Design Elements

When you are designing your document, it is just as important to know your audience when you are planning your document's text. Use of elements creates a sense of unified document. Headings and lists help readers note different

sections. Writers use design elements to its comprehension and keep reader's interest. Proper use of page layout- the writers make decisions about arranging white space, text, headings and graphics on their chosen medium to create a visually appealing reading experience.

White Space: whitespace is blank space with no text or graphics. The area in a design that is free of text, images, or other embellishments (lack of content).

- Space between words: the empty space between words in a sentence.
- Space around a picture: the empty space around a picture on a wall.
- Art museums: uses whitespace to provide a background for the paintings.
- Margins on a page: the empty space around the edges of a page.
- Visual rest: whitespace provides a break for the viewer's eyes, making the subject more interesting. White space can increase readability by up to 20%. It can help readers distinguish between different pieces of information and prevent eye fatigue.

Text: text refers to the words printed on the page. Readers the text more quickly if it is left- justified with right- edges; meaning that the text is floss with the left margin of the page and uneven along the right margin.

Type fonts, are divided into Serif and Sans Serif. Georgia is an example of Serif. Arial is an example of Sans Serif.

Serif is a decorative line small horizontal and vertical planes within a word while Serif is simple, clean lines that are of the same width. Serif fonts lend a traditional feel to everything from logos to business cards to marketing brochures are used in books, newspapers, and maganizes'titles. Serif is all about embracing tradition, Sans Serif fonts are about throwing the tradition out of the window in favour of modern sophistication and is considered more modern because they are minimal and simplistic.

Serif refers to letters with distinguishing lines or tails but sans Serif refers to letters with few distinguishing lines and no tails. Sans Serif fonts are appropriate for something sort, such as a title or heading.

Serif mixes capital and lower-case letters. Documents in all caps are difficult to read. Hence, all caps should be used for a title or major headings. For online

material, use a serif font in 16 points or higher. 10 or 12 is a comfortable reading size for most people.

SANS SERIF FONTS	POINT SIZE
Arial	16
Calibri	14
Verdana	12
Century Gothic	10

SERIF FONTS	POINT SIZE
Garamond	16
Times New Roman	14
Georgia	12
Courier	10

Once you consider the size; print styles such as boldface, underline, italics, and capital letters are to be placed. Don't overuse them, may distract the reader drawing attention away from your message.

Another tool for focusing readers attention is an itemized list. To itemize a list, you can use Bullets or Numbers. Some may use Alphabets or Romans.

Headings: earrings are short titles that introduces the main idea of a text. Adding help your readers see the organization of a document in one glance. Larger documents need headings followed by sub-headings with their page numbers.

Graphics: graphics are visual representation of information. They include familiar visual aids such as tables, line graphs, pie charts, bar graphs, figures and diagrams. Sophisticated aids such as schematic, pictographs, Internet graphics are used. Information can be expressed better in a visual form then in words alone. Graphics is often combined with text to convey meaning.

Medium: in addition to whitespace, headings and graphics, writers choose the best medium for their documents. Possible choices include paper media or electronic media- e-mail, social media, or web app.

Who Reads Graphics?

A picture is worth a thousand words. Graphics can clarify information quickly. At one glance, your readers can perceive more information that they would with words alone. Most complex technical material can be simplified with a graphic: a table, a drawing, or diagram, or a graph. Academic readers rely heavily on words do you understand meaning; Technical writers often rely on words as well as on graphics.

Audience: technical subjects such as engineering, marketing, and medicine rely heavily on data presented visually. Some concepts cannot be relayed without a graphic. Therefore, readers of technical documents expect to see graphic aids in their reading. While international readers may have difficulty with some vocabulary, they can often comprehend graphic quickly. Graphics depend on what your readers can understand and what they already know.

Generally, the more data in a graphic, the more difficult it is to read. When choosing which graphic your audience will best understand, ask these questions:

- How much does my reader know the subject?
- How interested is my reader in the subject?
- Do my readers need or expect technical information expressed visually as graphics? Or will my audience be confused by graphics?
- Does my audience's reading level tend to be higher or lower?
- How will my reader access the graphic, through traditional or online media?

Purpose and Objectives: audience is only one consideration when selecting a graphic. You must consider the purpose of your graphic as well as how much and what kind of information you want to convey. In order to fulfill the purpose, ask yourself these two questions:

- What is the purpose of the writing?
- How can graphics help achieve that purpose?

Then choose the graphic that paste meets your needs. To help select an appropriate graphic come on look at the purpose column table and decide what do you want to accomplish.

PURPOSE GRAPHIC

present a small amount of numerical data	informal table
Present numbers in easy-to-read rows and columns	formal table
Explain an idea using words	verbal table
Contrast data to highlight differences	bar graph and pictograph
Contrast 2 or more sets of data to highlight differences	multiple bar graph
Follow or trend over time	line graph
Compare 2 or more trends$ show how data is related	multiple line graph
Illustrate how the all is divided into parts or how the parts relate to the other	pie graph for chart
educate quickly with an attractive combination of tests and visuals	information graphics
explain a process	flow chart
demonstrate a decision-making process	decision flow chart
present the structure of an organization	organizational flow chart
confirm a schedule of task	Gantt/Grant chart
demonstrate an idea using a symbol	icon
illustrate representative details of an object or a mechanism	diagram or line drawing
show how something actually looks	photo

Designing Graphics

Graphics add critical information to your text and presence overall appearance of a document. Good writers create a flow between words and graphics that unifies or document allowing readers to move along without interruption. To help your reader interpret graphics quickly and easily, you should keep graphics neat and simple; integrate graphics with text; Give credit for borrowed graphics; and use color effectively.

a. Keep Neat and Simple: the quality of a graphic is always depended on how carefully it is read. On neat and clean graphic is easy to read and

interpret. So, leave enough white space around the graphic so the page looks organized, and make the graphic large enough so the readers can see all parts clearly. Don't use clotted graphics- untidy graphics. Graphic is nearly on readable with too many colored lines; it looks more like a robotic weapon from a science fiction movie then a line graph showing the trends.

Provide keys to measurements: explain how your data is measured and what the data means. There are different units of measurement for different types of measurements such as- second, minutes, hours days, weeks, years, metres kilometers, millimetres, centimetres, kilograms, litres, tons, centigrade, Fahrenheit. Distance, volume, are time should be considered because they are important in graphic design as they make quantities meaningful. In graphic design, even the slightest deviation from the intended measurements can result in errors in printing and binding, or significant visual discrepancy in online application. Whether it's creating a print design, or wave layout or a logo adheres to clients' requirements. Accurate measurements also contribute to the overall consistency and coherence of a design, which are crucial factors in maintaining a strong brand identity. Engage the viewers. Use symbols or colors for quick guides.

b. Integrate Graphics with Text: clearly refer to his graphic by figure or table number in the text before you place it on the page and as soon as your reader needs to look at it. Use the word figure to refer to any graph that is not a formal table. Use the word table to refer to a formal table- A table with rules and titles. No more figures consecutively throughout. Number tables consecutively- bought separately from figures- throughout. Call your readers' attention to figures or tables by incorporating references in your text, using parentheses, or creating stand-alone sentences as follows:
    - Table 1/ Figure 1 shows the amount of rainfall in Nepal over the past three years.
    - The rainfall in the valley over six years varied substantially. (See Table 1/ Figure 1).

Choose one method and refer to all your figures and tables the same way. Use the conventions for capitalization and abbreviations of your discipline. Informal

tables (a few numbers set apart from the text) need not be identified as a table or a figure. Verbal tables use words instead of numbers, and the information is presented in rows and columns.

Place is graphic in a convenient place for the reader to see. If the graphic is small enough (1/5 to ½ page), try to place it on the same page as its reference. If the graphic is large (3/4 to 1 page), you may need to place it on the page immediately following your reference. In reports with complex graphics, the graphics are sometime placed in an appendix, especially if the graphics interrupt the flow of the report. For graphics used in reports come on explain the significance of the graphic in the report as you introduce the graphic. For example, point out parts of a diagram, trends in a line graph, or important numbers in a table. You need not discuss every part of the, what do point out what is important for the reader to know. Generally, explanations are presented in the introduction preceding the graphic. More complex graphics may require discussion before and after placement of the graphic. Provide a title for every graphic, except for informal tables that are simply indented from the margin. Table titles and figure titles are often centered below the figure. However, placement may vary. Choose a placement and be consistent throughout the document. Use specific names for titles so readers understand what they are to learn from the graphic.

c. Give credit for borrowed graphic: Give credit for a graphic if you do not compile it yourself or if you compiled it using borrowed data. Place the word *source* below your figure. Include the bibliographical reference for the source as you would a footnote or an endnote and specify if you have permission to reprint the graphic (e.g., “Reprinted with Permission”) or if you have changed the graphic in some way (e.g., adapted from Williams, 2016, p.64”).
d. Use color effectively: color often attracts or readers’ attention before the words attract his attention. Color is a powerful design tool that can be used to:
    - Indicate a document’s organization: By using the same color for major headings, you give your readers visual cues to the overall organization of a document and help them scan quickly for information.

- Emphasize or clarify an important point: when using a color for a keyword, a phrase, an idea, or part of a graphic, you draw your readers' attention to important information. You also can use cooler to signify or change or to guide readers through a process.
- Support your text's meaning: By using certain colors consistently to convey specific information, your reader begins to associate the color with the meaning (red with danger) and therefore understands the text more quickly.
- Make your document attractive: By using colors to break up black-and-white text, your page will look more appealing.
- Improve visibility for visually impaired readers: Use contrasting colors to enable visually impaired readers to see your color scheme. Comply with ADA regulations by avoiding reds and greens together for readers who are red/green colorblind.

With a click of the mouse, you can change color from blue to green to red. But be careful. Like any technical writing tool, color can be misused. To use color effectively:

  - Avoid overusing color: More than three or four colors (including black and white as two colors) can overwhelm a reader, and using one color (other than black) too often can be distracting.
  - Apply color consistently to elements throughout a document: For example, if you set major headings in red and set key words in black boldface, do not suddenly switch a major heading to blue or a key word to red boldface.
  - Remain sensitive to cultural identifications with colors: International audiences associate colors differently from American audiences. For instance, Americans associate green with go (traffic lights) and red with danger (fire trucks and stop signs). However, green is a holy color to a Muslim and red is a sign of mourning to a South African.
  - Avoid unusual combinations of colors: Some colors, such as purple on a blue background and orange on red, are difficult to read.

Constructing Graphics

As a technical writer selecting graphics for a document, you have many options from which to choose. You make those choices by considering which graphics are most appropriate for different audiences and purposes.

a. Constructing tables: tables are used to present words or numbers organized into categories of rows and columns. Tables, are one of the most important popular graphic aids, can be informal or formal. A verbal table, often called a chart is a formal table, categorizing words instead of numbers into columns and rows.
   - Informal table: an informal table is a graphic that uses rows and columns drawn without lines or stubs (column headings). In an informal table, the information flows with the text, and is used to convey simple information.

   - Formal table: a formal table is an arrangement of information is rows and columns with rules (lines drawn). It is typically used to organize numbers, words, or other data in a consistent format. Numerical information, in particular can be difficult to follow without the structure of a table.

   - Verbal Table: Barwell table, also known as a chart, is similar to a formal table with its rows and columns. Like a formal table, it uses stubs and may include footnotes and bibliographic information if taken from another source. Formal tables use numerals; verbal tables use words.

b. Constructing Graphs:

Bar Graph: a paragraph can be defined as a graphical representation of data, quantities, or numbers using bars or strips. They are used to compare and contrast different types of data, frequencies, or other measures of distinct categories of data. It is a way to display data graphically using rectangle bars. These bars are uniform in width but vary in height or length. Also, the length of each bar represents a value. In addition, the bars can be vertical or horizontal. We use bar graphs to compare various items choices. We also use them to show how something has changed over a period of time. They can also be used to show the

frequency of certain data. Other names for bar graphs are bar charts or column graphs.

Divided Bar Graph: a divided bar graph refers to a type of bar chart where each bar is further subdivided into sections, visually representing the proportions of different categories within a single whole, essentially showing how a total amount is broken down into its component parts; it is also sometimes called a “sub divided bar graph” or “component bar chart.”

Multiple Bar Graph: A multiple bar graph is a graph that has multiple bars of each category. A multiple bar graph compares as many sets of data you want. The process for creating a multiple bar graph is just like creating any other bar graph, only you’ll have more colors to represent different sets of data. A multiple bar graph shows the relationship between different values of data. Each data value is represented by a column in a graph.

Histogram: is a bar graph representation of data. The horizontal axis displays the number. The vertical axis shows frequency amount of data present in each range.

Line Graph: line graph is similar to a bar graph that it uses or horizontal axis and vertical axis to compare numerical data. Instead of bars, however, this graph uses a line to show the trends.

Multiple Line Graph: create a multiple line graph to illustrate the data. First, draw the horizontal line and the vertical axes. Next, label the vertical axis. The vertical axis lists the dependent variable and represents the quantity of the data. Usually, multiple line graphs are used to show trends over time. In the graph, each data value is represented by a point in the graph that are connected by a line. In a multiple line graph, there are two more lines in the graph connecting two or more sets of data. The independent variable is listed along the horizontal, or X axis and the quantity or value of the data is listed along the vertical, or Y axis.

Pictograph: the pictorial representation of data using images, icons, or symbols. We can represent the frequency of data using symbols or images. Pictographs add color and interests, are easy to read, and are especially appropriate for nonnative speakers of English. To be clear, the pictures or icons must be easily recognized and distinguishable from one another.

Information Graphic: information graphics, or infographics, are visual representations of data that use images, charts command and text to communicate information. They are used to make complex data easy to understand. They are used in newspapers to display whether, or in the maps.

c. Charts:

Pie Chart: a pie chart is a type of graph representing data in a circular form, with each slice of the circle representing a fraction or proportionate part of the whole.

Flowchart: a flow chart is a type of diagram that represents a workflow or process. It is a graphical representation of steps. A flow chart is a diagram that shows how the parts of a system work together. It is used to visualize the processes, inputs, outputs, and decisions that make up a system.

Organizational chart: A diagram that visually conveys a company's internal structure by detailing the roles, responsibilities, and relationships between individuals within the organizations. It is a way to convey an organization's structure highlighting the different jobs, departments, and responsibilities that connect the company's employees to each other and to the management team. Organizational charts graphically display and employee's hierarchical status relative to other individuals within the company.

Gantt chart: A Gantt chart is a bar chart that illustrates a project schedule. It was designed and popularized by Henry Gantt around the years 1910-1915. A Gantt chart is a project management tool assisting in the planning and scheduling of projects of all sizes; they are particularly useful for visualizing projects.

d. Others:

Diagram: a drawing, sketch, plan, or chart that makes something clearer or easier to understand. Labels in diagrams are used to show the different parts.

Icon: As a pictogram, an icon is a graphical representation of an object, place or idea. An icon is a small picture or symbol that represents a larger image.
Photograph: a picture made using a camera, in which an image is focused on to light-sensitive material and then made visible and permanent by chemical treatment, or stored digitally. The process and practice of creating such images is called photography.